AMA515A Forecasting and applied time series analysis

Ch.2 Stationarity and further exponential smoothing

2.1 Time series

A stochastic process is a family of random variables Z_t, where t belongs to the set of integers $0, \pm 1, \pm 2, \ldots$. If t carries the meaning of time, then the process is called a time series. In most cases, we shall be dealing with non-negative time, namely, t = 0, 1, 2,….

2.1.1 Stationary time series

A strictly stationary series is one whose statistical properties remain constant in time. A proper definition for a time series $\{X_t, t = 0, \pm 1, \ldots\}$ is by the condition that $(X_1, \ldots, X_n)$ and $(X_{1+h}, \ldots, X_{n+h})$ have the same joint distribution for all integers h and n > 0. A necessary condition for this is the absence of any deterministic or stochastic trend. Stationary series are amenable to formal analysis. Elimination of non-stationarity is a primary aim of preliminary, descriptive analysis.

2.1.2 White noise

If $X_t \sim IID(0, \sigma^2), \forall t$, the process $\{X_t\}$ is said to be white noise. It's a term coming from engineering. Thus white noise is the simplest kind of strictly stationary time series. Before attempting any analysis in terms of more complicated models, it is advisable to examine, either informally or by formal tests, whether data show any incompatibility with white noise. A standard tool is the autocorrelation coefficients in the next section.

2.1.3 Weakly stationary TS

Strict stationarity is usually a too strong condition in applications. Let $\mu_t = E(X_t), \sigma_t^2 = E(X_t - \mu_t)^2$ and $\text{cov}(t, t+h) = \gamma(t, t+h)$. Here cov is the abbreviation of covariance. If

(i) μ_t is independent of t, and
(ii) $\gamma(t, t+h)$ is independent of t for each h,

then X_t is said to be weakly stationary time series. In this course we mainly focus on this type of stationarity.

2.2 Further discussion on weak stationarity

Recall if a discrete time stochastic process $\{X_t\}$ satisfy the following conditions:

(i) $E(X_t) = \mu$ (constant) for all t
$\text{var}(X_t) = \sigma^2$ (constant) for all t

(ii) $\text{cov}(X_t, X_s) = \Gamma_{t-s}$ (depend on t-s only !)

Then $\{X_t\}$ is said to possess second order stationarity. Some authors also say $\{X_t\}$ is weakly stationary or covariance-stationary.

2.2.1 Autocorrelation function of a stationary stochastic process

The autocovariance function of a time series X_t is defined by

$\gamma(k) = E(X_t X_{t+k}) - E(X_t)E(X_{t+k})$, and $\rho(k) = \frac{\gamma(k)}{(\text{var}(X_t)\,\text{var}(X_{t+k}))^{1/2}}$. Here $k = 0, \pm 1, \pm 2, \ldots$.

For a stochastic process which is stationary, then the autocovariance function is simplified to

$\gamma(k) = E(X_t X_{t+k}) - E(X_t)^2$ and $\gamma(0) = \text{var}(X_t)$.

In this case, the autocorrelation function $\rho(k) = \frac{\gamma(k)}{\gamma(0)}$, k=0, ±1 , ±2, …

Remark 1 : It can be shown that $|\rho(k)| \le 1$.

Remark 2: Some algebraic simplicity can be obtained if $E(X_t)=0$, because then $\gamma(k) = E(X_t X_{t+k})$ and $\gamma(0) = \text{var}(X_t) = E(X_t^2)$. But remember this is a special case. In general $E(X_t) \neq 0$ and we have to use the full form above.

Example 1:

Let $\{a_t\}$ be zero-mean white noise and $\text{var}(a_t) = \sigma^2$. A time series is formed by the following process:

$Z_t = a_t + 0.5a_{t-1}$

Show that Z_t is stationary by finding its autocovariance function.

Solution: $E(Z_t)=0$, $\text{var}(Z_t)=\sigma'^2$ (constant).

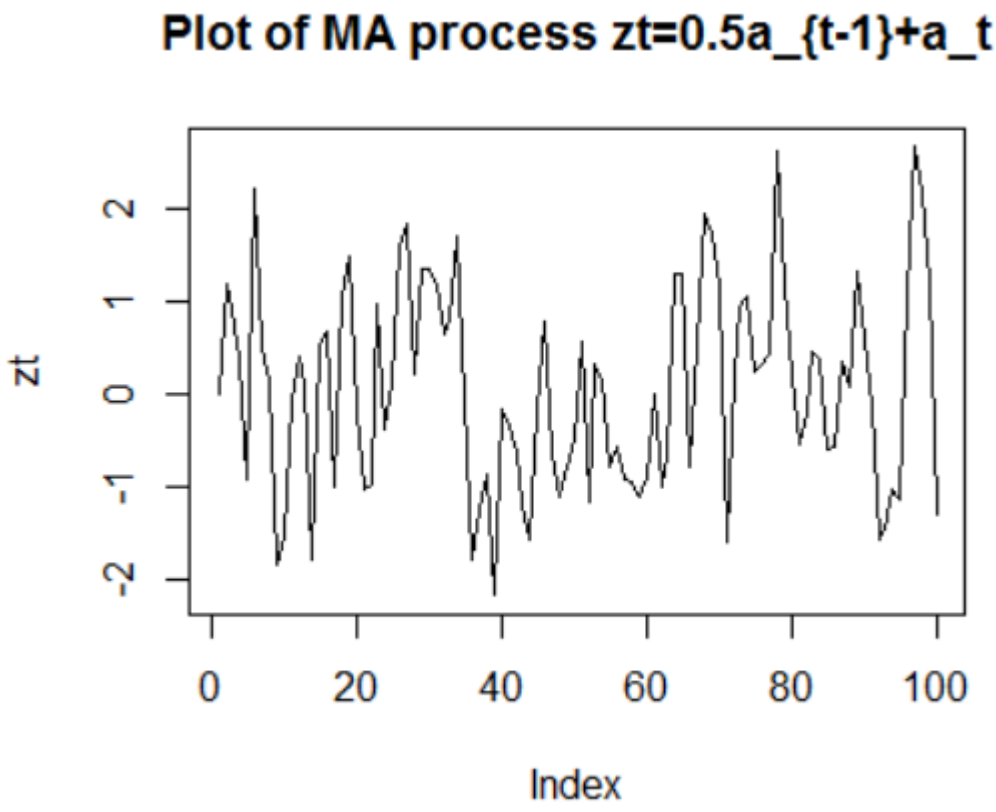
Observe $E(Z_t Z_{t-k}) = E(a_t+0.5a_{t-1})(a_{t-k}+0.5a_{t-k-1})$
$= E(a_t a_{t-k} + 0.5a_{t-1}a_{t-k} + 0.5a_t a_{t-k-1} + 0.25a_{t-1}a_{t-k-1})$

$$= \begin{cases} 1.25\sigma^2, if\ \ k=0 \\ 0.5\sigma^2, if\ \ k=1 \end{cases}$$

and zero otherwise. Thus the autocovariance function does not depend on t and the time series is stationary.

Data set of Z_t:

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[1]  0.000000000  1.199029059  0.937211975  0.377127620 -0.924223292
2.220568702  0.527257800
  [8]  0.142674172 -1.834390395 -1.528993473 -0.085945082
0.416225614  0.121972651 -1.782135004
 [15]  0.534948229  0.679576935 -0.996696720  1.069333629
1.490594600 -0.191680799 -1.015607750
 [22] -0.966119524  0.982169064 -0.364061477  0.126410622
1.595840358  1.842386935  0.222502768
 [29]  1.366782692  1.375200233  1.194991685  0.657327092
0.808684808  1.714032542 -0.057458359
 [36] -1.788004908 -1.247961109 -0.868670219 -2.173200872 -
0.159922316 -0.341009101 -0.628610581
 [43] -1.175171658 -1.555893585 -0.182386914  0.802749687 -
0.626516274 -1.091144944 -0.779253637
 [50] -0.490859319  0.583532012 -1.170744961  0.334446867
0.182804797 -0.784265916 -0.549941369
 [57] -0.916257097 -0.943788382 -1.109368599 -0.878730597 -
0.001349007 -1.007794331 -0.661527550
 [64]  1.296500949  1.296215840 -0.786050349  0.544770321
1.968043939  1.741360250  1.167194140
 [71] -1.589614677  0.205395416  0.961343896  1.050667001
0.252366853  0.320838385  0.427223073
 [78]  2.643445289  1.190330743  0.200806139 -0.536378755 -
0.180027191  0.479469412  0.389866555
 [85] -0.591259974 -0.549441985  0.348061201  0.094776612
1.323005303  0.612284727 -0.125294291
 [92] -1.558313190 -1.467395135 -1.027625129 -1.118878250
1.055524758  2.680406656  2.186709476
 [99]  1.233137154 -1.285679126
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Example 2:
Suppose $Y_t = 5 + 2t + X_t$, where X_t is a zero-mean stationary series with autocovariance function γ_k .

(a) Find the mean function for Y_t.
Solution: $E(Y_t) = 5 + 2t$ as X_t has mean zero.

(b) Find the autocovariance function for Y_t.
Solution: $E(Y_t - E(Y_t))(Y_{t-k} - E(Y_{t-k}))$
$= E(5+2t+X_t-5-2t)(5+2(t-k)+X_{t-k}-5-2(t-k))$
$= E(X_t X_{t-k}) = \gamma_k$, the same autocovariance function as X_t.
(c) Is Y_t stationary?

Example 3:

Consider the TS $Z_t = A\sin(\omega t + \theta)$,
where A is rv with a zero mean and unit variance and θ is rv with a uniform distribution on the interval $[-\pi, \pi]$ independent of A. Then

$$E(Z_t) = E(A)E[\sin(\omega t + \theta)] = 0$$

$$E(Z_t Z_{t+k}) = E[A^2 \sin(\omega t + \theta)\sin[\omega(t+k)+\theta]]$$
$$= E(A^2)E\left\{\frac{1}{2}[\cos(\omega k) - \cos(\omega(2t+k)+2\theta)]\right\}$$

$$= \frac{1}{2}\cos(wk) - \frac{1}{2}E\left\{\cos\omega(2t+k)+2\theta)\right\}$$

$$= \frac{1}{2}\cos(\omega k) - \frac{1}{2}\int_{-\pi}^{\pi}\cos(\omega(2t+k)+2\theta)\frac{1}{2\pi}d\theta$$
$$= \frac{1}{2}\cos(\omega k) - 0$$
$$= \frac{1}{2}\cos(wk)$$

Thus Z_t is stationary.

2.2.2 Sample Autocorrelation

For a series $\{X_t : t = 1,...,n\}$, define

$$C_k = \sum_{t=k+1}^{n} X_t X_{t-k} - (n-k)\bar{X}^2, k = 0,1,...$$

Then

$$r_k = \frac{C_k}{C_0}$$

is the kth sample autocorrelation coefficient of $\{X_t\}$. A plot of r_k vs k is called the correlogram of $\{X_t\}$

Notes:

i) Limits $\pm 2/\sqrt{n}$ are conventionally used to assess significantly non-zero r_k

ii) Slowing decaying r_k, especially with $r_1 \approx 1$, is indicative of non-stationarity in $\{X_t\}$. May need to eliminate trend.

iii) Sudden peaks in r_k suggest seasonal effects, again eliminate wherever possible.

iv) Removal of trend or seasonal effect can induce autocorrelation pattern in the residual series

v) Let X_t be a time series, then the autocovariance function of X_t is defined as

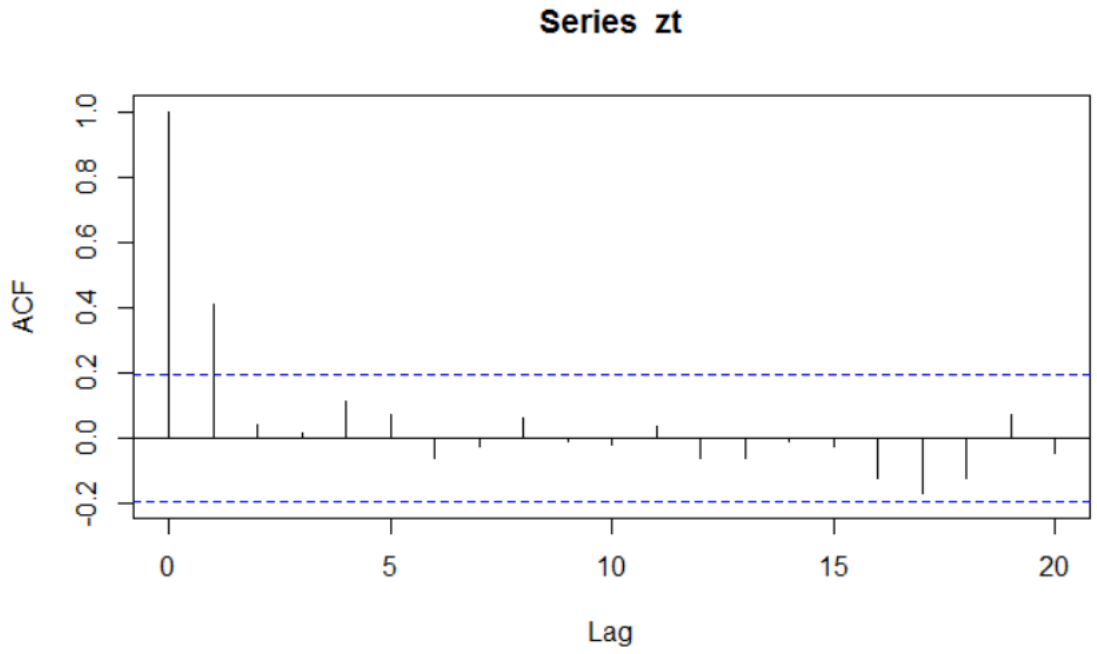
$$Cov(X_t, X_{t-k}) = E(X_t - E(X_t))(X_{t-k} - E(X_{t-k}))$$

The autocorrelation function of X_t is defined as

$$Corr(X_t, X_{t-k}) = Cov(X_t, X_{t-k})/\{\text{var } X_t \text{ var } X_{t-k}\}^{1/2}$$

Example 4:

Consider the MA1 process Z_t of Example 1, then its sample autocorrelations is shown below. Notice the theoretical lag-1 autocorrelation coefficient is 0.4, while the sample autocorrelation coefficient is slightly larger than 0.4 as shown in the plot.



2.3 General Linear Process

A general linear process (GLP) $\{Y_t\}$ is one that can be represented by a linear combination of present and past white noise terms, namely

$$Y_t = \psi_0 e_t + \psi_1 e_{t-1} + \ldots = \sum_{i=0}^{\infty} \psi_i e_{t-i}$$

Without loss of generality, ψ_0 is usually set to be 1. For the right hand side of the series to be convergent, we need conditions on the ψ_i terms. For our purpose, we impose the condition $\sum_{i=1}^{\infty} \psi^2{}_i < \infty$.

Theorem: A general linear process with condition $\sum_{i=1}^{\infty} \psi^2{}_i < \infty$ is stationary.

Proof: By direct calculation, $E(Y_t)=0$, and $\text{var}(Y_t) = E(Y_t^2) = \sigma^2{}_e \sum_{i=0}^{\infty} \psi^2{}_i < \infty$, free of t .

Finally $\text{cov}(X_t, X_{t+k}) = E(X_t X_{t+k}) = E(\sum_{i=0}^{\infty} \psi_i e_{t-i} \sum_{j=0}^{\infty} \psi_j e_{t+k-j}) = (\sum_{i=0}^{\infty} \psi_i \psi_{i+k}) \sigma^2{}_e$,

which does not depend on t.

2.4 Exponential smoothing for a linear trend process

Suppose our process is of the form

$$x_t = b_1 + b_2 t + \varepsilon_{t,} \quad \text{where} \quad \varepsilon_t \sim NID(0, \sigma^2),$$

if we use simple exponential smoothing, then recall

$$S_T = \alpha \sum_{k=0}^{T-1} \beta^k x_{T-k} + \beta^T S_0 \text{, where } \beta = 1 - \alpha$$

$$\Rightarrow E(S_T) = \alpha \sum_{k=0}^{T-1} \beta^k E(x_{T-k}) + \beta^T S_0$$

$$= \alpha \sum_{k=0}^{T-1} \beta^k [b_1 + b_2 (T-k)] + \beta^T S_0$$

As T goes to ∞, we have

$$E(S_T) = (b_1 + b_2 T)\alpha \sum_{k=0}^{\infty} \beta^k - b_2 \alpha \sum_{k=0}^{\infty} k\beta^k$$

$$= b_1 + b_2 T - (\beta/\alpha) b_2 = E(x_T) - (\beta/\alpha) b_2 \qquad (2.4.1)$$

Thus the smoothed statistic is a biased forecast for a linear trend process.

2.5 Double exponential smoothing (for linear trend process)

Recall simple exponential smoothing, i.e.

$$S_T = \alpha x_T + (1-\alpha) S_{T-1},$$

is a biased forecast for linear trend process. Now if we let

$$S_T^{[2]} = \alpha S_T + (1-\alpha) S_{T-1}^{[2]} \qquad (2.5.1)$$

it can be shown that

$$E(S_T^{[2]}) = E(S_T) - \frac{\beta}{\alpha} b_2 \qquad (2.5.2)$$

thus

$$b_2 = (\alpha/\beta)[E(S_T) - E(S_T^{[2]})]. \qquad (2.5.3)$$

A logical estimate of b_2 at the end of time T is thus

$$\hat{b}_2(T) = \frac{\alpha}{\beta}(S_T - S_T^{[2]}) \qquad (2.5.4)$$

Using (2.4.1) and (2.5.3), we observe

$$E(x_T) = E(S_T) + \frac{\beta}{\alpha}\frac{\alpha}{\beta}\left[E(S_T - E(S_T^{[2]})\right]$$

$$= 2E(S_T) - E\left(S_T^{[2]}\right) \qquad (2.5.5)$$

Hence a reasonable estimate for x_T at the end of time T is

$$\hat{x}_T = 2S_T - S_T^{[2]} \qquad (2.5.6)$$

Now

$$\hat{b}_1(T) + \hat{b}_2(T).T = \hat{x}_T$$

$$\Rightarrow \hat{b}_1(T) = 2S_T - S_T^{[2]} - T.\frac{\alpha}{\beta}(S_T - S_T^{[2]}) \qquad (2.5.7)$$

Initial estimates satisfy

$$S_0 = \hat{b}_1(0) - \frac{\beta}{\alpha}\hat{b}_2(0)$$

$$S_0^{[2]} = \hat{b}_1(0) - 2.\frac{\beta}{\alpha}\hat{b}_2(0)$$

where $\hat{b}_1(0)\ and\ \hat{b}_2(0)$ are usually obtained by regression analysis on historical data.

Forecasting

If we always use a 'current origin approach', then we have

$$\hat{x}_T(\tau) = \hat{a}_1(T) + \tau\hat{b}_2(T),$$

where $\hat{a}_1(T) = \hat{x}_T = 2S_T - S_T^{[2]}$, by (2.5.5)

$$\because x_{T+\tau} = b_1 + b_2(T+\tau)$$

$$= b_1 + b_2T + b_2\tau$$

$$= E(x_T) + b_2\tau$$

Rem 1: Rationale behind d. exp. smoothing is that S_T lags behind x_T by the 'same' amount that $S_T^{[2]}$ lags behind S_T. Thus we add back to S_T the amount S_T-$S_T^{[2]}$, i.e.

$$\hat{x}_T(\tau) = b_1 + b_2(T+\tau) \text{ theoretically}$$

$$= E(x_T) + \tau b_2$$

$$\approx S_T + (S_T - S_T^{[2]}) + \tau\hat{b}_2(T), \text{ by (2.5.6)}$$

$$= 2S_T - S_T^{[2]} + \tau\frac{\alpha}{\beta}(S_T - S_T^{[2]})$$

$$= S_T + (1+\tau\frac{\alpha}{\beta})(S_T - S_T^{[2]}) \qquad (2.5.8)$$

Thus $E(\hat{x}_T(\tau)) = b_1 + b_2(T+\tau)$, showing the forecast is unbiased.

Note: I use the symbol $\hat{x}_T(\tau)$ to denote the τ-step ahead forecast made at time T

Choice of smoothing constant

The smoothing constant controls the number of past realizations of the TS that influence the forecast. Small values of the smoothing constant give weights to many prior observations and result in a slow response to changes to the TS.

However, large smoothing constant respond more rapidly to the change in the TS, it may cause the system to respond to random variations, when actually the system (TS) does not change.

Some authors suggest the rule that we choose between 0.01 and 0.3 for a constant process.

A quantitative method is to carry out a sequence of trials on a set of historical data using several different values for the smoothing constant, and then to select an α that optimizes some measure of effectiveness. One approach is using Least Squares. We consider

$$f(\alpha) = \sum_{t=m}^{m+k} \left\{ x_{t+1} - \alpha \sum_{j=0}^{m-1} (1-\alpha)^j x_{t-j} \right\}^2$$

and minimize f (α) numerically to find α.

2.6 Higher order smoothing (General polynomial)

The d. exp. Smoothing method can be extended easily to higher order. For example, if the model is :

$$x_t = b_1 + b_2 t + (1/2) b_3 t^2 + \varepsilon_t ,$$

with current origin forecasting equation

$$\hat{x}_{T+\tau}(T) = \hat{a}_1(T) + \hat{a}_2(T)\tau + (1/2)\hat{a}_3(T)\tau^2 ,$$

then it can be shown

$$\hat{a}_1(T) = 3S_T - 3S_T^{[2]} + S_T^{[3]}$$

$$\hat{a}_2(T) = \frac{\alpha}{2\beta^2}\left[(6-5\alpha)S_T - 2(5-4\alpha)S_T^{[2]} + (4-3\alpha)S_T^{[3]}\right]$$

$$\hat{a}_3(T) = \left(\frac{\alpha}{\beta}\right)^2 (S_T - 2S_T^{[2]} + S_T^{[3]})$$

where

$$S_T = \alpha x_T + (1-\alpha)S_{T-1},$$

$$S_T^{[2]} = \alpha S_T + (1-\alpha)S_{T-1}^{[2]}, \text{ and}$$

$$S_T^{[3]} = \alpha S_T^{[2]} + (1-\alpha)S_{T-1}^{[3]}.$$

The above procedure is known as triple exponential smoothing. Result can be generalized to n^{th} degree polynomial directly.

An Example

Four years of monthly data (1995-1998) for the sales of a kind of filter are available. We'll use it to illustrate the calculations of the D. Exponential Smoothing method. We use the first two years (1995, 1996) data to find the initial estimates (Regression method), and show the forecasting for the year 1997. The table below gives the results:

Month of 1997	Period	Actual sales	S_T	$S_T^{[2]}$	$\hat{x}_{T-1}(1)$
Jan	1	538	446.28	352.22	541
Feb	2	570	458.65	362.86	551
March	3	600	472.79	373.85	565
Apr	4	565	482.01	384.67	583
May	5	485	482.31	394.43	590
June	6	604	494.48	404.44	580

Note : $\hat{x}_{T-1}(1)$ **is the one-step ahead forecast for time T made at T-1**

Calculations :

Assume S_0 and $S_0^{[2]}$ are known from the regression based on 1995 and 1996.
Let S_0 and $S_0^{[2]}$ be 436.09 and 341.77, and let the smoothing constant α be 0.1.

Recall the forecasting equation, (2.5.8), we have

$$\hat{x}_0(\tau) = S_0 + (1+\tau\frac{\alpha}{\beta})(S_0 - S_0^{[2]}),$$

thus the 1-step ahead forecast at time 0 (Dec. 1996) is

$$436.09 + (1+\frac{0.1}{0.9})(436.09-341.77) = 540.89 \approx 541$$

At the end of Jan. 1997, we know the sales for that month is 538 units. To calculate the forecast for Feb., we have to find S_1 *and* $S_1^{[2]}$ first, they are

$S_1 = 0.1(538)+(1-0.1)436.09=446.28$
$S_1^{[2]}=0.1(446.28)+(1-0.1)(341.77)=352.22$

Thus $\hat{x}_1(1) = 550.79 \approx 551$, by (2.5.8) again. The other forecasts are obtained in a similar fashion.